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Industrial Automation

Course AutB/AAut/ADP, *select one and delete others*

Author 1: first name.last name
Author 2..3... if any.

Lab title...

Goal of the lab

Your summary in some lines

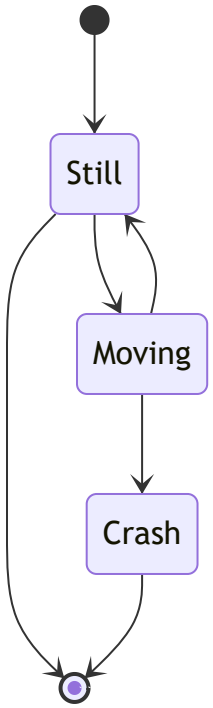
1. ....
2. ....

Form of document

1. Text documents in **.md** format. See MD Basic Syntax as reference
2. Any text script between tags, e.g.
   i. `iecst` for IEC 61131-3 structured text.
   ii. `js` for javascript.
3. Any image in the **img** folder.
4. Any other document than md file or image in **documents** folder.
5. Any other page with link to the page.

6. Any diagram, i.e. UML, state machine or other in mermaid form. Do not forger `mermaid tags`

Finally: zip all documents with last name of all authors as `zip_name.zip` and upload them in cyberlearn, *by mail if problems with cyberlearn*.



Length of the document

About 150 to 200 lines, check your vs-code editor.

Core of the document

Select the most relevant points and summarize them if necessary with text and diagrams.

About code

- Quality of IEC 61131-3 code is as important as functionnality.

  `It is always possible to make bad code work if you dedicate enough time to it.`
- For POU code, it is better to separate core and header.

Header

```
VAR
    ijStArray   : ARRAY [1..GVL_ARRAY_SIZE.I_MAX_SIZE] OF stArrayOfDint;
END_VAR
```

Core

```
FOR iMyLoop := 1 TO GVL_ARRAY_SIZE.I_MAX_SIZE BY 1 DO
    FOR jMyLoop := 1 TO GVL_ARRAY_SIZE.J_MAX_SIZE BY 1 DO
        ijStArray[iMyLoop].jArray[jMyLoop] := iMyLoop * jMyLoop;
    END_FOR
END_FOR
```

Tests

- Summarize what you have done, eventually not done and why, e.g. time.
- What you have teste.
- What is working.

e.g.

Subject	Action	Test	Result
Open gripper	Code written	Push button	Gripper opens
Code for FB_O300_DL	Code written	not done because of time	failed
Axis regulation	Modified parameters	Quality of regulation	Oscillations
...	...	...	...

What you have learned

In one sentence.

What you did not understand.

In one sentence.